

The Volume Dividend

Energy density, blend dilution, and who profits per extra litre — ethanol and isobutanol in petrol, biodiesel in diesel, versus compressed biogas in CNG

India, FY 2024-25 volume basis · July 2026 prices · companion to the OMC retail-profitability model

Executive summary

Ethanol carries 34% less energy per litre than petrol. Blending it dilutes every litre sold, cutting km-per-litre — yet the pump price per litre is unchanged, so the consumer silently buys more litres for the same distance. Because central excise, state VAT (in effect), dealer commission and OMC marketing margin are all levied per litre, every stakeholder in the fuel chain collects on the extra volume. At E20 on India's 54-billion-litre petrol pool, the volume effect alone moves ≈₹22,700 crore per year of extra consumer spend: ≈₹8,500 crore to the exchequer, ≈₹886 crore to retail-outlet dealers, ≈₹757 crore to OMC marketing margin. Walking the blend to E30 roughly doubles this.

Compressed biogas (CBG) is the structural exception: purified to IS 16087 spec it matches or exceeds pipeline CNG in energy per kilogram, and CNG is retailed per kg — so blending CBG into the gas stream substitutes imports without any hidden dilution levy on the consumer. It is, fiscally, the honest blend.

1. Energy content of the fuels (lower heating value)

Fuel	MJ/kg	MJ/litre	Sold by	vs base
Petrol (E0)	43.4	32.1	litre	—
Ethanol (anhydrous)	26.8	21.1	litre	−34%
Isobutanol	33.1	26.5	litre	−17%
Diesel (B0)	43.0	35.7	litre	—
Biodiesel (FAME)	37.2	32.7	litre	−8%
HVO renewable diesel	44.0	34.3	litre	−4% (drop-in)
CNG (pipeline gas)	47.5	n/a (gas)	kg	—
CBG (IS 16087, CH ₄ ≥90%)	46.5	n/a (gas)	kg	≈ CNG parity
Pure methane (CBG ceiling)	50.0	n/a (gas)	kg	+5% vs CNG

The oxygenates that policy favours for petrol are exactly the ones with the deepest energy deficit. Isobutanol — denser, less oxygenated — halves ethanol's penalty per unit of renewable content. On the gas side the ordering flips: well-purified biogas sits at or above fossil CNG.

2. Blend dilution → extra litres for the same distance

Blend	MJ/L	Energy vs base	Extra litres (energy basis)	Real-world drop
E10	31.02	−3.4%	+3.5%	~2%
E20 (today)	29.92	−6.8%	+7.3%	4% (SIAM 2–6%)
E25	29.37	−8.5%	+9.3%	~5.5%
E30	28.82	−10.3%	+11.4%	~7%
IB16 isobutanol	31.22	−2.8%	+2.9%	no India data
IB24 isobutanol	30.78	−4.2%	+4.3%	no India data

Blend	MJ/L	Energy vs base	Extra litres (energy basis)	Real-world drop
B7 biodiesel	35.48	-0.6%	+0.6%	≈ energy basis
B10 biodiesel	35.39	-0.8%	+0.8%	≈ energy basis
B15 biodiesel	35.25	-1.2%	+1.3%	≈ energy basis
B20 biodiesel	35.10	-1.7%	+1.7%	≈ energy basis
CBG in CNG (any %)	per kg	0%	0%	0% — sold per kg

Real-world E20 loss (4%) runs below the pure energy math (6.8%) because E20-calibrated engines claw back part of it through octane and combustion gains; older vehicles lose up to ~12%. Diesel has no equivalent recovery mechanism, so biodiesel's real-world drop tracks the energy arithmetic.

3. What the dilution costs a petrol owner

Hatchback, 20 km/L on E0, 10,000 km/yr, pump ₹105/L. The price board shows the same ₹/litre for E0 and E20 — the energy cut is invisible at the point of sale:

Blend	Extra litres / yr	Extra ₹ / yr	Hidden levy on every km
E20	+21 L	₹2,188	4.2%
E25	+29 L	₹3,056	5.8%
E30	+38 L	₹3,952	7.5%

4. National petrol pool: where the extra litres' money goes

Base: FY24-25 petrol pool 54.1 bn L (40 MMT), distance demand held at its E0-equivalent, dispensed through 99,281 retail outlets. Per-litre stakes: excise ₹19.90, effective state VAT ≈₹19.5, dealer commission ₹4.1, OMC margin ₹3.5. All figures ₹ crore per year:

Blend	Extra bn L	Consumer pays	Central excise	State VAT	Dealer comm.	OMC margin
E20	2.16	22,703	4,303	4,216	886	757
E25	3.02	31,712	6,010	5,889	1,238	1,057
E30	3.91	41,011	7,773	7,616	1,601	1,367
IB16*	1.49	15,696	2,975	2,915	613	523
IB24*	2.22	23,295	4,415	4,326	910	776

*Isobutanol rows are energy-basis (no Indian fleet-trial data). The OMC-margin column reconciles exactly with the OMC retail-profitability model (₹757 / 1,057 / 1,367 crore).

4a. The E20 → E30 walk: what each next step adds

India is already at ~E20, so the live policy question is the increment. Deltas below are each blend's take beyond what E20 already extracts:

Step (vs E20)	Δ bn L	Δ Consumer	Δ Excise	Δ VAT	Δ Dealer	Δ OMC
E25	+0.86	+9,009	+1,707	+1,673	+352	+301
E27	+1.30	+13,621	+2,581	+2,530	+532	+454
E30	+1.75	+18,308	+3,470	+3,400	+715	+610

The walk from E20 to E30 roughly doubles the volume-effect take. The increments are also qualitatively worse than the first 20%: today's engines are calibrated for E20, so the octane-recovery offset that softened E20's

real-world drop has little left to give at E25–E30 — each further step passes through closer to the raw energy loss.

4b. Diesel pool: B7 → B20 biodiesel walk

Diesel is the bigger prize by volume — 110 bn L/yr (91.4 MMT), twice the petrol pool. FAME dilutes far less per litre, but the base is enormous. Per-litre stakes: pump ₹92, excise ₹15.80, effective VAT ≈₹12.3, dealer ₹3.1, OMC ₹2.5:

Blend	Mileage drop	Extra bn L	Consumer pays	Excise	VAT	Dealer	OMC
B7	−0.6%	0.64	5,904	1,014	786	199	160
B10	−0.8%	0.92	8,455	1,452	1,126	285	230
B15	−1.2%	1.38	12,736	2,187	1,696	429	346
B20	−1.7%	1.85	17,053	2,929	2,271	575	463

B20 pulls nearly as many extra litres as E20 does on petrol, though the per-km burden is far gentler (−1.7% vs −4%). The incidence differs too: diesel is freight-dominated, so the extra cost cascades into logistics and inflation rather than household fuel budgets. Feasibility check: actual biodiesel blending was under 1% in FY24–25 and the National Biofuel Policy targets only B5 by 2030 — B15/B20 are a what-if ceiling, and OEM warranties beyond B7 remain unresolved.

5. The CBG contrast: same decarbonisation, no hidden levy

- Petrol + ethanol — energy dilution is real; pump ₹/L unchanged; consumer buys ~4% more litres at E20; centre, state, dealer and OMC all collect per-litre on the extra volume.
- Diesel + FAME — the same mechanism, mild per litre but material in aggregate because the pool is 2× petrol's.
- CNG + CBG — retailed per kilogram at methane-grade energy (46.5 MJ/kg spec floor vs 47.5 pipeline; pure methane 50). Substituting CBG changes sourcing — domestic agricultural and municipal waste instead of imported LNG — without changing the kilograms bought per kilometre. Zero dilution, zero volumetric pass-through.

Both routes substitute imports. Only the liquid route monetises an energy haircut through per-litre levies. CBG delivers the import substitution without taxing the consumer through the fuel gauge — and its SATAT procurement pricing keeps the subsidy explicit and on-budget rather than hidden in mileage.

6. CBG economics under SATAT pricing

Anchoring mileage on the SIAM/ARAI BS-VI fuel-efficiency declarations (303 models): petrol 16.67 kmpl, diesel 17.91 kmpl, CNG 27.40 km/kg — and, decisively, the same nameplate travels +40.3% further per unit on CNG than on petrol (11 factory petrol/CNG pairs).

6a. What a megajoule costs to procure

Source	₹/MJ	Basis
Petrol (refinery / trade parity)	1.81	₹58/L
Ethanol (OMC procurement)	2.94	₹62/L
Domestic APM gas	0.51	~₹24/kg CNG-eq
Imported spot RLNG	0.94	~₹45/kg CNG-eq (\$12/MMBtu)
CBG (SATAT assured)	1.16	₹54/kg ex-plant + 5% GST

Ethanol's renewable premium over the petrol it displaces is +63% per MJ — before the mileage-dilution levy on the consumer. CBG's premium over the spot RLNG it displaces is +23% per MJ, and it undercuts imported gas outright whenever spot LNG runs above ~\$14.8/MMBtu. Per megajoule of renewable energy bought, SATAT CBG (₹1.16) is 2.5× cheaper than ethanol (₹2.94), with no volumetric side effects.

6b. Cost per km on SIAM/ARAI declared FE (Delhi prices)

Fuel	₹/km	Mileage basis
Petrol E0	6.30	16.67 kmpl (SIAM declared)
Petrol E20	6.56	E20 real-world -4%
Petrol E30	6.77	E30 real-world -7%
Diesel B0	5.14	17.91 kmpl (SIAM declared)
Diesel B20	5.23	B20 energy basis -1.7%
CNG	2.74	27.40 km/kg (SIAM declared)
CBG (any % in CNG)	2.80	energy-scaled 46.5 / 47.5 MJ/kg

CNG runs at ~43% of petrol's cost per km — the +40.3% distance uplift compounding the per-unit price gap. The blend walk moves petrol the wrong way (₹6.30 → ₹6.77/km from E0 to E30) while CBG blending leaves the CNG ₹/km essentially untouched: even 100% CBG at the IS 16087 purity floor costs 2.1% in mileage, and at obligation-level shares it rounds to zero.

6c. The CBG Blending Obligation: national cost, zero consumer levy

CBO mandates 1% CBG in CGD gas from FY25-26, stepping toward ~5% by FY28-29. On the 6.67-MMT CNG(T) pool, at SATAT ₹54/kg (+5% GST) versus spot RLNG ~₹45/kg as the displaced marginal molecule:

CBO share	CBG needed (t/yr)	Procurement (₹ cr/yr)	Pump-price delta	Mileage delta
1% (FY25-26)	66,700	378	+₹0.12/kg	-0.02%
5% (FY28-29)	333,500	1,891	+₹0.59/kg	-0.11%

Even the full 5% obligation moves the CNG pump by ~₹0.59/kg (~0.8%) and mileage by ~0.1% — against E20's silent 4% mileage cut and ₹22,700 crore/yr volume dividend. Where CBG does need support, the subsidy sits on-budget and ex-plant — SATAT price assurance, GOBARDhan capex grants, and fermented-organic-manure offtake under FCO Schedule VIII — rather than hidden in the fuel gauge. Caveat: ₹54/kg is an assured floor, not proof of plant viability; feedstock economics (press-mud cheapest, agri-residue dearest) still decide whether plants get built.

7. Price-adjustment scenarios: what would neutralise the mileage loss

If blended petrol were priced for honest cost-per-km, its pump price would carry a discount equal to the mileage it takes away: $P(\text{blend}) = P(\text{E0}) \times (1 - \text{drop})$. At a ₹105 pump, that means:

Blend	Mileage drop	Parity discount	Parity pump price	National cost (₹ cr/yr)
E20	4.0%	₹4.20/L	₹100.80	22,703
E25	5.5%	₹5.78/L	₹99.22	31,712
E30	7.0%	₹7.35/L	₹97.65	41,011

Note the FY23 budget's ₹2/L penalty on unblended petrol is a stick pointing the other way — it widens the per-km gap instead of closing it. Four funding routes, tested against the ethanol-procurement, grain-price and taxation numbers from the companion models:

S1 — pass through the tax break already embedded (works, 1.3–1.5× coverage)

From the Bang-for-Your-Buck tax analysis: the ethanol molecule escapes both central excise (₹19.90/L) and state VAT (~₹19.5/L), bearing only 5% GST (~₹3.1/L on ₹62 ethanol) — but because E20 sells at the unblended price, that saving is retained in the price build-up rather than rebated. Netting off ethanol's cost premium over refinery petrol (₹62 vs ₹58):

Blend	Embedded headroom (₹/L)	Parity needs (₹/L)	Coverage
E20	6.46	4.20	1.54×
E25	8.07	5.78	1.40×
E30	9.69	7.35	1.32×

The embedded tax break more than funds parity at every blend level. Passing it through prices E20 honestly with ~₹2/L still left in the chain — no new subsidy and no new revenue loss versus an unblended counterfactual. What it does end is the windfall from the volume effect: that gain is, litre for litre, the consumer's loss.

S2 / S3 — centre or states fund it via duty cuts (possible, costly)

An excise cut of the full parity discount leaves ₹15.70/L (E20) falling to ₹12.55/L (E30) and costs the centre ₹22,700–41,000 crore/yr; the state-VAT route is symmetrical. Either is politically the mirror image of today's arrangement — which is precisely why neither has happened.

S4 — fund it from cheaper ethanol (fails on grain economics)

Parity funded from procurement alone needs ethanol at ₹41/L (E20) down to ₹37.5/L (E30) — far below every ESY 2024-25 feedstock slab:

Feedstock	OMC procurement (₹/L)
C-heavy molasses	57.97
FCI surplus rice	58.50
B-heavy molasses	60.73
Damaged food grains	64.00
Sugarcane juice / syrup	65.61
Maize	71.86

Grain and cane mandi prices set these floors — and maize, the marginal feedstock the E30 walk leans on, is the most expensive slab, pushing average procurement up, not down. Squeezing procurement to the cheapest slab funds only ₹0.81/L of E20's ₹4.20 need (~19%). Ethanol economics cannot pay for parity; only the tax side can. CBG needs no such scenario at all — its energy parity per kilogram is physical, not fiscal.

8. Ethanol supply: can the distilleries serve the blend walk?

Methodology. Demand per blend is computed from this paper's own pool arithmetic — the FY 2024-25 blended petrol pool (54.05 billion litres) is held at constant distance-demand, each blend's real-world mileage drop re-inflates the litres dispensed, and the ethanol fraction of those litres is the fuel-ethanol requirement. To this we add non-fuel ethanol demand (potable liquor and industrial chemicals). Supply is taken from three independent sources: (i) CareEdge Ratings — the credit-rating agency formerly CARE Ratings — in its 14 May 2026 study 'E85 Impact: Ethanol Overcapacity to Persist as Flex-fuel Transition to Remain Gradual'; (ii) the sanction register of

the Department of Food & Public Distribution (DFPD) Ethanol Interest Subvention Scheme (ISS), extracted in the companion E20→E30 stakeholder workbook; and (iii) the National Cooperative Development Corporation (NCDC) disbursement split for the Ministry of Cooperation's Cooperative Sugar Mill (CSM) strengthening scheme, as reported to the Rajya Sabha.

Supply-side item	Value	Source
Installed distillery capacity, ESY* 2025-26	~2,000 crore litres/yr	CareEdge May-2026
Additions expected operational by FY27	+400 crore litres/yr	CareEdge May-2026
DFPD ISS sanction register (approved, not all built)	1,212 projects · 1,37,282 KLPD† = 4,530 cr L/yr	DFPD annexures
Ethanol offered that OMCs currently absorb	~60%	CareEdge May-2026
Non-fuel demand (potable + industrial)	300–350 crore litres/yr	CareEdge May-2026
FCI‡ rice leg, effective	~211 crore litres/yr (≈3.9 blend points)	7.2 MMT/yr allocation × 65% lifting × 450 L/tonne
State skew	Maharashtra +277 / Tamil Nadu –77 cr L	CareEdge May-2026

*ESY = Ethanol Supply Year (November–October). †KLPD = kilolitres per day; annualised at 330 operating days. ‡FCI = Food Corporation of India, which sells surplus rice to distilleries under the Open Market Sale Scheme (Domestic).

Result — the higher blends are the cure for the overcapacity, not a casualty of it. Capacity utilisation (fuel + non-fuel demand ÷ capacity) by blend, on today's 2,000 crore litres and on the 2,400 available by FY27:

Blend	Fuel ethanol (cr L)	Total incl. non-fuel	Utilisation on 2,000	On 2,400 (FY27)	Verdict
E20 (today)	1,081	1,406	70%	59%	structural overcapacity — CareEdge's ~60% absorption
E25	1,373	1,698	85%	71%	mid consolidation band
E27	1,494	1,819	91%	76%	sweet spot — absorbs state surpluses, no new build
E30 (today's pool)	1,674	1,999	100%	83%	fits; leans on marginal maize
E30 demand-grown, FY30-31	2,173	2,498	—	~104%	sanction register must actually build

Two subsidiary findings. First, the cooperative-sugar-mill 'wave' is a working-capital rescue, not a capacity wave: of the ₹10,005 crore the NCDC disbursed to 56 cooperative sugar mills, 96.5% (₹9,657 crore) went to working capital, ₹97 crore to cogeneration, and only ₹251 crore to ethanol plants — which at the DFPD scheme's average intensity (₹96.8 crore of loan per 113-KLPD project) buys roughly 293 KLPD ≈ 9.7 crore litres a year, 0.5% of installed capacity. India's 229 functional cooperative mills (~30% of sugar output) remain marginal to an ethanol build-out that is in fact a private, grain-based wave (Uttar Pradesh alone added ~50 crore litres of capacity in 2025). Second, the Food Corporation of India rice leg is small and politically fragile: ~211 crore litres effective (at a 65% lifting assumption against the 7.2-million-tonne allocation), about 3.9 blending points, and it carries the July-2023 precedent when rice-to-ethanol was suspended outright — so the E27/E30 increment leans on open-market maize, which is also the dearest procurement slab (₹71.86/L). Feedstock, not distillation steel, is the binding constraint.

Assumptions and caveats: capacity figures are nameplate, not effective (maintenance, feedstock switching and working-capital stress reduce them); the 65% FCI lifting rate is a lever (FY26 all-channel offtake experience, band 55–75%); non-fuel demand is held flat at the CareEdge midpoint (325 cr L); the FY30-31 row uses this repo's petrol-demand forecast base case (5.0% CAGR, moderating on electric-vehicle penetration); and CareEdge's infrastructure caveat transfers intact — a single-grade retail network of ~1.03 lakh outlets, ~77.8 crore litres of storage and ~300 blending depots suits one national blend stepped E20→E27, but not a multi-grade E85/flex-fuel world.

9. The RON95 octane dividend: the credit the consumer never sees

RON — the Research Octane Number — measures a fuel's resistance to knock (uncontrolled combustion), which is what limits an engine's compression ratio and therefore its thermal efficiency. Indian regular petrol is specified at RON 91 (Bureau of Indian Standards IS 2796); premium grades (XP95, Speed 95) at RON 95 sell for roughly ₹8–10/L more. Ethanol is a poor energy carrier but an excellent octane carrier: its blending RON is ~112 (neat RON ~108; the blending value is higher because ethanol's effect in a mixture is super-linear). Every blend step therefore delivers an octane credit alongside its energy debit — and where that credit lands is a policy choice.

Methodology. We evaluate the two limiting routes with a linear blending model: route (a), today's practice, holds the finished pump fuel at RON 91 and solves for the blendstock-for-oxygenate-blending (BOB — the petrol base the refinery actually makes) the refiner needs; route (b) holds the BOB at RON 91 and computes the pump RON the blend delivers. The octane credit is priced two ways: at the retail premium-grade spread (~₹2.25 per RON point, an upper bound reflecting willingness to pay) and, as the caveat notes, at refining cost (₹0.3–0.8/L per 4–5 points, the resource-cost lower bound).

Blend	(a) BOB needed if pump stays RON 91	(b) Pump RON if BOB stays 91	Octane credit at retail spread (₹/L)
E10	88.7	93.1	4.7
E20	85.7	95.2	9.5
E25	84.0	96.2	11.8
E27	83.2	96.7	12.8
E30	82.0	97.3	14.2

Route (a) — today — means that at E20 the refinery only has to make ~85.7-RON blendstock, five points below the old specification: lower reforming severity, more cheap naphtha in the pool, a saving retained upstream on top of the tax headroom of Section 6. Route (b) is the opportunity: the identical E20 on an unchanged 91-RON base is a RON-95 fuel — premium-grade octane, nationally, at zero incremental refining cost, and the blend walk carries it to RON 97.3 at E30.

Why this matters for mileage: a RON95-labelled E20 lets manufacturers raise compression ratios (about 10.5 → 12, feasible within the knock limit at RON 95). At the standard ~1.5% thermal-efficiency gain per compression-ratio point, that is +2.2% efficiency against E20's ~4% energy drop — a net penalty of ≈ –1.8% on an engine designed for the fuel. This is Brazil's actual playbook: E27 regular petrol at ~RON 95+, high-compression engines, no headline mileage complaint. It composes cleanly with Section 7: parity pricing compensates the existing fleet immediately, while RON95 labelling plus E20-plus engines make the penalty physically shrink as the fleet turns over (~2.9 crore new registrations a year; the Vahan registry already tracks PETROL(E20) as its own fuel type, 21% of CY2025 registrations).

Assumptions and caveats: octane blending is non-linear and blendstock-dependent — 112 is a central literature value (range 108–115) and the linear model is an estimate; the retail-spread valuation of the credit is a willingness-to-pay ceiling, not a refining cost; the compression-ratio gain arrives only with new, RON95-calibrated engines, not by relabelling the fuel for the existing parc; and compressed biogas needs no octane accounting at all — methane's octane rating is ~120+, and CNG engines already run compression ratios above 12.

10. DME in LPG: the fourth quadrant

Dimethyl ether (DME, CH₃-O-CH₃) is now blendable into LPG up to 20% under BIS standard IS 18698:2024, positioned as an import-substitution play for a fuel India imports ~60–65% of. The engineering evidence comes from LERC — the LPG Equipment Research Centre, Bengaluru, the oil-industry body that certifies LPG equipment: material compatibility cleared at 20% DME in the vapour phase, flex-fuel burner trials stable, and a measured thermal-efficiency decrease of 5.26% at DME20 on the IS 4246 burner protocol. That 5.26% is the cooking-gas analogue of SIAM's 4% E20 mileage figure — and, like it, runs below the pure energy arithmetic (–7.4%), because oxygenated DME burns cleaner and claws part of the deficit back.

Methodology. DME carries 28.8 MJ/kg against LPG's 45.8 (Indian 60:40 butane:propane mix) — a 37% energy deficit per kilogram, deeper than ethanol's 34% per litre. We blend by mass, take LERC's measured drop as the central usable-energy case (scaled linearly to DME5/10), and apply the same constant-demand re-inflation as the petrol analysis: households cook the same, so they buy more kilograms. LPG's two structural differences drive the result: it is sold per kg in a fixed 14.2-kg cylinder, and domestic LPG is subsidised (PMUY, ₹300/cylinder to ~10.3 crore beneficiary households).

Blend	MJ/kg	Energy vs LPG	LERC-basis drop	Cylinder energy (MJ)	Extra kg for same cooking
DME5	44.95	-1.9%	-1.3%	638	+1.3%
DME10	44.10	-3.7%	-2.6%	626	+2.7%
DME20	42.40	-7.4%	-5.3%	602	+5.6%

Who pays, at DME20 on the full 29.7-MMT pool: a 7-cylinder household buys +0.39 cylinders = ₹312/yr at an unchanged ₹803 price; nationally ₹9,325 crore/yr of extra consumer spend — and, the inversion, ≈₹1,201 crore/yr of extra PMUY subsidy. Ethanol's dilution earns the exchequer excise on the extra volume; DME's dilution bills it, because the diluted fuel is subsidised. The four-quadrant summary of this paper's whole argument:

Blend	Dilution	Fuel's fiscal status	The volume effect lands on
Ethanol in petrol	-34% /L	heavily taxed	consumer pays; exchequer + trade collect
Biodiesel in diesel	-8% /L	taxed	mild; cascades into freight
CBG in CNG	~0% /kg	lightly taxed	nobody — the honest blend
DME in LPG	-37% /kg	subsidised	consumer AND exchequer pay

The economics do not clear either: energy-neutral DME must price at ≤₹32.7/kg (LPG import parity ₹52 × the energy ratio), but methanol-route DME costs ₹33.6/kg in feedstock alone at imported methanol — swapping a propane import for a methanol import at a worse ₹/MJ. Import substitution is honest only on domestic-carbon routes (coal-gasification methanol at ₹15–18/kg, or bio-DME), and DME20 would need 6.3 MMT/yr of DME — roughly six times India's methanol output. If DME blending proceeds, two guardrails follow from this paper's framework: energy-parity cylinder pricing (₹761 for a DME20 cylinder at today's ₹803, or equivalently more kilograms per cylinder), and domestic-carbon DME only.

Assumptions and caveats: LERC's 5.26% is a burner thermal-efficiency figure at DME20; field behaviour and older stoves may differ, and the linear scaling to DME5/10 is an estimate. IS 18698:2024 caps blends at 20% — beyond it, DME attacks conventional elastomer seals and LERC's compatibility work is the binding reference. The whole-pool blend is a ceiling scenario; a phased rollout scales linearly. Cylinder price (Delhi ₹803), PMUY parameters and LPG LHV (45.5–46.3 by season mix) are levers in `dme_lpg_blending.py`.

11. Caveats and sources

- Vehicle mileage: SIAM/ARAI BS-VI FE declarations (form 2344, 303 four-wheeler models, Apr 2020) from the `vehicle_fuel_mileage` repo — type-approval figures, used for relative fuel-type gaps; the +40.3% petrol→CNG uplift is anchored on 11 same-nameplate pairs.
- CBG/SATAT: MoPNG assured ex-plant price ₹54/kg (+5% GST); CBO 1% of CGD gas FY25-26 → ~5% FY28-29; gas comparators APM \$6.5/MMBtu, spot RLNG \$12/MMBtu at ₹83/\$.
- Volumes: PPAC Ready Reckoner FY2025-26 (H1), Table 6.1 (MS 40.0 MMT, HSD 91.4 MMT FY24-25); outlets Table 6.6/6.7 (99,281); dealer commission Table 8.10.
- Mileage drops: SIAM/ARAI central figures (E20 4%, band 2–6%; up to ~12% for older vehicles); E25/E30 scaled on ethanol's calorific deficit; isobutanol and biodiesel rows are energy-basis.
- Per-litre levies are editable levers shared with `omc_model.py` and `statewise_tax_impact.py`: excise ₹19.90/L petrol, ₹15.80/L diesel; VAT 25% on ₹78/L petrol base, 17.5% on ₹70/L diesel base; OMC margins ₹3.5/₹2.5.

- This volume effect is distinct from — and additive to — the tax-differential effect (states forgo VAT on the ethanol fraction, ~₹9,000 cr/yr at E20): states lose on substitution while every per-litre stakeholder gains on volume.
- Supply side: CareEdge Ratings, 'E85 Impact: Ethanol Overcapacity to Persist as Flex-fuel Transition to Remain Gradual', 14 May 2026; DFPD Ethanol Interest Subvention Scheme annexures (via the E20→E30 stakeholder workbook); NCDC/Ministry of Cooperation CSM-scheme disbursements (Rajya Sabha reply); FCI rice allocations from the FCI-warehouse repo (OMSS(D) 2024-25/2025-26 orders); Uttar Pradesh capacity from the PARIVESH environmental-clearance register (digital-twin layer 24d).
- DME side: LERC (LPG Equipment Research Centre, Bengaluru) DME-LPG compatibility and IS 4246 thermal-efficiency work (~5.26% at DME20); BIS IS 18698:2024 (20% blend cap); PPAC LPG consumption; PMUY subsidy parameters.
- RON side: BIS IS 2796 fuel specification; ethanol blending-RON literature (108–115, central 112); retail XP95/Speed-95 price spreads; compression-ratio/efficiency literature (~1.5% per CR point).
- Generated by build_energy_doc.js from energy_blend_comparison.py / price_parity_scenarios.py / cbg_satat_economics.py / ethanol_supply_match.py / ron_octane_analysis.py outputs; do not hand-edit. Repository: github.com/herrickshaw/india-omc-fuel-fleet-model.

Annex A — The CBG incentive stack

Compiled from the official GOBARDhan important-circulars register (gobardhan.eil.co.in, 70 circulars as of July 2026), cross-checked against the primary gazette for the fertiliser leg (CG-DL-E-29032025-262118). The stack de-risks every leg of the CBG business model — feedstock, capital, debt, gas revenue, digestate revenue, carbon revenue — and every rupee of it sits on-budget and ex-plant, the structural contrast with ethanol's hidden consumer-side funding documented in this paper. Roughly half the register is dated 2025-26: the policy cadence is accelerating.

Lever	Instrument (issuer, date)	What it provides
Offtake & pricing	SATAT assured procurement (MoPNG, May-2022; methodology & price revision May-2025)	OMCs guarantee CBG offtake at an assured ex-plant price (₹54/kg basis, excl. 5% GST) under long-term agreements
	CBG-CGD Synchronization Scheme (MoPNG, Apr-2021; extended Oct-2023, Mar-2026)	CBG injected into city-gas networks, fungible with natural gas — grid sale instead of trucking
	CBG Blending Obligation (MoPNG, Feb-2024)	Mandatory 1% CBG in CNG(T)/PNG(D) from FY25-26, stepping toward ~5% by FY28-29 — demand by decree
Capital support	MNRE Waste-to-Energy / National Bioenergy Programme (Nov-2022; revised Jun-2025; suppl. budget Sep-2025)	Central financial assistance per unit of CBG capacity, capped per project
	SBM 2.0 bio-methanation VGF (MoHUA, cost norms revised Nov-2025)	Viability-gap funding for municipal-waste CBG plants
	SASCI 2026-27 (DoF Mar-2026; MoPNG guidelines May-2026)	Special capital assistance to states tied to CBG-sector reforms — the newest money in the register
Infrastructure & feedstock	Pipeline Infrastructure (DPI) scheme (MoPNG, 2024; amended Jan-2025, revised Aug-2025, extended Feb-2026)	Funds the connecting pipeline from plant to the CGD grid — the offtake bottleneck
	Biomass Aggregation Machinery scheme (MoPNG, Feb-2024; revised Jul-2025)	Assistance for paddy-straw / biomass collection machinery
	Crop Residue Management guidelines (DA&FW, 2023-24)	Channels crop stubble to CBG plants as feedstock
Fertiliser (digestate)	FCO amendments: FOM Jul-2020, LFOM May-2021,	Digestate becomes a notified fertiliser;

Lever	Instrument (issuer, date)	What it provides
leg	2023 relaxations → Schedule VIII 'Organic Carbon Enhancer' (DA&FW, Feb/Mar-2025)	Schedule VIII defined on organic carbon (12 – 14%) + heavy metals only — no NPK/C:N/sieve, built so CBG digestate passes
	Bulk Sale Notifications I–VII (DA&FW, Nov-2023 → Feb-2026)	Named plants may sell FOM/LFOM in bulk directly to farmers for 3 years, bypassing dealer licensing
	Market Development Assistance (DoF, Sep-2023 + SOP/iFMS/DBT machinery 2024-25)	~₹1,500/tonne support on FOM/LFOM/PROM sold — a per-tonne digestate subsidy
	ICAR Package of Practice (DARE/ICAR, Jul-2024)	Official agronomic guidance legitimising FOM/LFOM use on crops
Finance access	Priority Sector Lending (RBI, Sep-2020)	CBG plants qualify for mandated, cheaper bank credit
	Agriculture Infrastructure Fund (Oct-2020; guidelines Jan-2023) · AHIDF (DAHD, Apr-2022)	Interest subvention + credit guarantee; dung-based plants eligible under the animal-husbandry fund
Tax & carbon	5% GST on CBG · excise exemption on CBG-blended CNG (Feb-2023 budget)	Light single tax; the blended portion is not taxed twice
	Carbon-credit eligibility (MoEFCC, Feb-2023)	CBG plants can monetise credits under the Paris-Agreement framework — a third revenue leg
Standards & regulatory easing	IS 16087:2025 Bio-CNG specification (Aug-2025) · MoRTH bio-CNG fuel notification (Jun-2015) · CPCB re-categorisation (2021, 2025)	Legal fungibility with CNG (the basis of energy parity per kg), automotive-fuel authorisation, friendlier pollution-control classification
State layer	UP 2022 · Bihar 2023/2025 · Rajasthan & AP 2024 · Assam 2025 · MP Apr-2025 · Chhattisgarh Jul-2026 · Model State CBG Policy (MoPNG, Apr-2026)	Capital-subsidy top-ups, land, stamp-duty and electricity-duty waivers; the model policy gives states a template

Caveats: scheme amounts (capital-assistance rates, MDA quantum) are periodically revised — the register dates above identify the governing circular to check; the NCDC cooperative-mill route nominally covers biogas-adjacent assets but is 96.5% working capital in practice (Section 8); and state-policy benefits vary widely in generosity and actual disbursement.

Glossary of abbreviations

Abbreviation	Meaning
LHV	Lower Heating Value — usable energy content of a fuel, excluding water-vapour condensation heat
RON / BOB	Research Octane Number (knock resistance) / Blendstock for Oxygenate Blending (the petrol base before ethanol is added)
E20, E25, E27, E30	Petrol blended with 20/25/27/30% ethanol by volume; B7–B20 likewise for biodiesel in diesel
FAME / HVO	Fatty-Acid Methyl Ester (conventional biodiesel) / Hydrotreated Vegetable Oil (drop-in renewable diesel)
CNG / CBG / CBO	Compressed Natural Gas / Compressed Biogas (purified to IS 16087) / CBG Blending Obligation on city-gas distributors
SATAT	Sustainable Alternative Towards Affordable Transportation — MoPNG scheme assuring CBG offtake at ₹54/kg ex-plant
OMC / RO	Oil Marketing Company (IOCL, BPCL, HPCL...) / Retail Outlet (petrol pump)

Abbreviation	Meaning
VAT / SGST / GST	state Value-Added Tax on petrol • State share of Goods & Services Tax • Goods & Services Tax (ethanol pays 5%)
DFPD / ISS	Department of Food & Public Distribution / its Ethanol Interest Subvention Scheme (2018-22 windows)
NCDC / CSM	National Cooperative Development Corporation / Cooperative Sugar Mill
FCI / OMSS(D)	Food Corporation of India / Open Market Sale Scheme (Domestic) under which FCI rice reaches distilleries
KLPD / cr L / MMT / ESY	Kilolitres per day • crore litres (10 million L) • million metric tonnes • Ethanol Supply Year (Nov-Oct)
SIAM / ARAI / FE	Society of Indian Automobile Manufacturers / Automotive Research Association of India / Fuel Efficiency
PPAC / RR	Petroleum Planning & Analysis Cell / its Ready Reckoner data book
RLNG / APM / MMBtu	Regasified Liquefied Natural Gas • Administered Price Mechanism (domestic gas) • Million British thermal units
FFV / CR / NBP	Flex-Fuel Vehicle • Compression Ratio • National Biofuel Policy
DME / LERC	Dimethyl Ether (blendable into LPG to 20% under IS 18698:2024) / LPG Equipment Research Centre, Bengaluru
LPG / PMUY	Liquefied Petroleum Gas / Pradhan Mantri Ujjwala Yojana (₹300-per-cylinder subsidy to ~10.3 cr households)